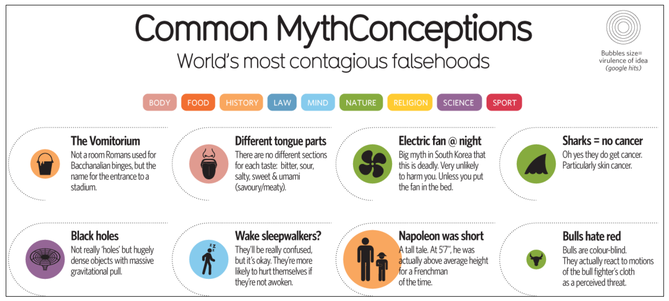
How Does it Happen?
A Moment-By-Moment Examination of Knowledge Revision

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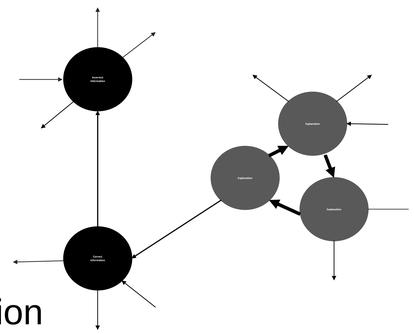
THE PROBLEM



KNOWLEDGE REVISION

The Knowledge Revision Components (KReC) Framework (Kendeou & O'Brien, 2014).

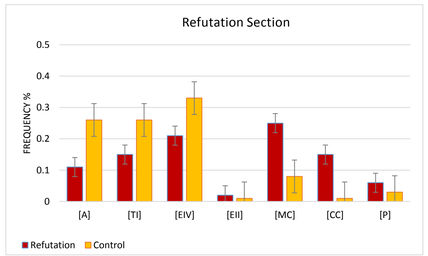
- Encoding
- Passive activation
- Co-activation
- Integration
- Competing activation

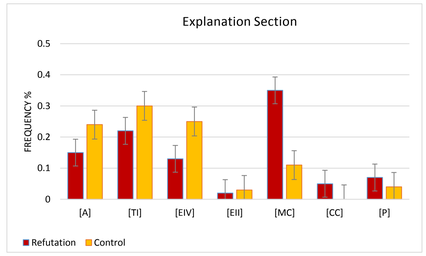


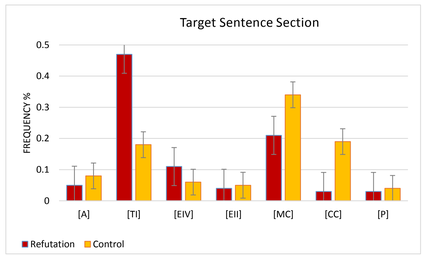
EVIDENCE

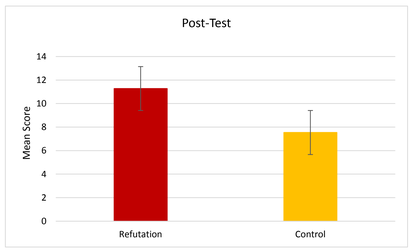
- Convergence between online (reading times, think-alouds, eye-tracking) and offline methodologies (post-tests, recalls).

Kendeou, van Boekel, & Quillien (2015)









Refutation texts resulted in the *co-activation and integration* of the correct and the incorrect conceptions (as indicated by cognitive conflict and comprehension monitoring processes). To the extent that the correct outcome sentence got integrated in the mental representation (as indicated by text-based and elaborate inferences), *the revision process was facilitated*.

Introduction
Tom loved all of his classes this year but science was still his favorite subject. His teacher Mrs. James enjoyed having Tom in class because he was very enthusiastic. She encouraged him to conduct his own simple experiments. After school one day Tom was testing something he learned that day. He had two balls that were exactly the same size and shape but one weighed twice as much as the other. He planned to drop both balls from the roof of his house at the same time. He wanted to see which one of the two balls would fall faster.

Refutation
Tom had initially believed that if two objects were the exact same shape and size but one was heavier, the heavier object would fall faster. His teacher explained that this idea was incorrect.

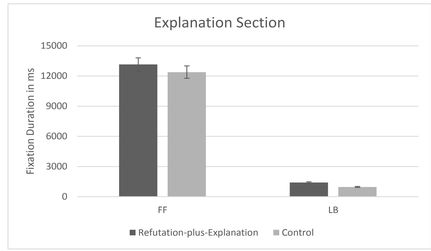
Explanation
She explained to Tom that when you drop two identically shaped objects that have different masses, the two objects will experience the exact same acceleration due to gravity. This is because gravity acts on all objects in exactly the same manner and this occurs regardless of their mass. It does not make a difference if one of the objects weighs more than the other object. That is to say, gravity will exert the same pull on both of them. Therefore the balls will fall at exactly the same rate. Tom's experiment was going to show that

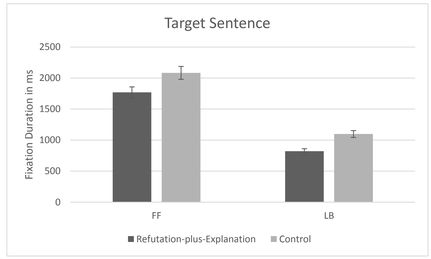
Target Sentence
the balls hit the ground at the same time.

Spillover Sentence
Tom was ready to conduct the experiment.

Closing
He was excited to tell his teacher about it the next day in school. He hoped she would be impressed and suggest other experiments that he could do at home.

Kendeou et al (2014)







More processing time (and likely attention) **to the causal explanation** during first pass but also during second pass reading.

This increased processing reflects *the construction of a highly interconnected causal network* that facilitates the processing (and *integration*) **of the correct outcome sentence.**

IMPLICATIONS - APPLICATIONS

- Design of Effective Texts - Messages
- Design of Technology-Based Learning Environments
- STEM Learning
- Teacher PD - Beliefs

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